Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/28/2017
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Fundamental Research on International Standard of Fire Safety in Space - Subteam 2: Wire Combustion with External Radiation in Support of the JAXA Project Fundamental Research on International Standard of Fire Safety in Space		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2012 Japanese Space Agency (JAXA) AO for Fundamental Research on an International Standard of Fire Safety in Space
Start Date:	03/06/2014	**End Date:**	03/05/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Prof. Carlos Fernandez-Pello is U.S. Co-Investigator on Japan Aerospace Exploration Agency (JAXA)-sponsored project, “Flammability Limits At Reduced-g Experiment (FLARE)." JAXA Principal Investigator is Prof. Osamu Fujita, Hokkaido University.		
COI Name (Institution):			
Grant/Contract No.:	NNX14AF01G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Funding is for Prof. Fernandez-Pello's role as U.S. Co-Investigator for the Japan Aerospace Exploration Agency (JAXA)-sponsored project, “Flammability Limits At Reduced-g Experiment (FLARE).” JAXA International Announcement of Opportunity (AO) to fund experiments to be conducted aboard the Japanese Experiment Module, Kibo, 2012. The objective of the proposed research program is to continue the experimental study of the flammability of wire materials in space exploration atmospheres and associated computational/theoretical tools to aid interpretation of test results.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying materials flammability in spacecraft allows us to accurately elucidate the effect of the environment parameters on the ignition and flame spread over combustible material, and through them their potential fire hazard. Particularly important is the determination of the Limiting Oxygen Concentration (LOC) on flame extinction under spacecraft environments. The anticipated improved methodology should reduce time and cost for the spacecraft material screening. The investigation and results have also benefits for terrestrial fire safety by providing further information about the flammability of materials under a variety of environments.
Task Progress:	A program to study the flammability characteristics of electrical wiring insulation materials that will complement the existing JAXA project entitled “Fundamental Research on International Standard of Fire Safety in Space” (FLARE) so as to provide an assessment of wire materials flammability in the conditions expected in “Space Exploration Atmospheres” (SEA) has been continued at University of California (UC) Berkeley under NASA sponsorship for the last year. A summary of the research conducted during the period March 1, 2016 to February 28, 2017 is reported here. The research has one primary goal: to understand the changes in electrical wire flammability, specifically the LOI-LOC (Limiting Oxygen Index-Limiting Oxygen Concentration), in the environment expected in space-based facilities, particularly those at exploration atmospheres (elevated oxygen concentration and low pressure). To date, flammability testing has focused mostly on atmospheric conditions. Work conducted at NASA Glenn Resarch Center (GRC) and at the combustion laboratories at UC Berkeley indicate that materials are more flammable in SEA atmospheres. However, considerable work remains to understand how the new cabin environment influences material flammability, particularly that of electrical wires. The project experimental methodology consists of obtaining flame spread flammability boundaries (LOI-LOC) of electrical wires and wire insulation materials (primarily those being studied in Hokkaido University) in SEA environments and under external radiation. To conduct the experiments, an experimental apparatus and a bench-scale flammability test were developed during the previous grant period. This apparatus was used during the present period to continue the studies of LOC of wire insulation for flame spread at different environmental conditions and external radiation. The testing methodology is supported with theoretical modeling of the ignition and flame spread process. While carrying out the flame spread tests it was observed that melting and dripping of the wire insulation had an important role in the process and that it influenced both the rate of flame spread and the LOC of the material. For this reason a subproject was initiated to study the role of wire insulation melting and dripping on the wire ignition and flame spread process. This is very significant because the burning characteristics of the wire insulation would be drastically different in normal and microgravity. Studies related to this subject comprised most of the work conducted during this reporting period. We feel that the findings to date are both novel and important to spacecraft fire safety. The research primarily supports the above mentioned JAXA program but also could lead to better determination and ranking of the fire hazard characteristics of potential wire insulation materials to be used in spacecraft.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
Abstracts for Journals and Proceedings	Kobayashi Y, Huang X, Fernandez-Pello C. "Melting and Dripping of Polyethylene Insulation in Flame Spread over Wires." 54th Symposium (Japanese) on Combustion, Sendai, Japan, November 23-25, 2016. Japanese Section of the Combustion Institute. 54th Symposium (Japanese) on Combustion, Sendai, Japan, November 23-25, 2016. , Nov-2016
Abstracts for Journals and Proceedings	Kobayashi Y, Konno Y, Huang X, Nakaya S, Tsue M, Fujita O, Fernandez-Pello C. "Flame Spread over Insulated Wires: the Role of Insulation Melting and Core Conductivity." 4th International FLARE Workshop, Noordwijk, Netherlands, January 2017. 4th International FLARE Workshop, Noordwijk, Netherlands, January 2017. , Jan-2017
Articles in Peer-reviewed Journals	Miyamoto K, Huang X, Hashimoto N, Fujita O, Fernandez-Pello C. "Limiting oxygen concentration (LOC) of burning polyethylene insulated wires under external radiation." Fire Safety Journal. 2016 Nov;86:32-40. http://dx.doi.org/10.1016/j.firesaf.2016.09.004 , Nov-2016
Papers from Meeting Proceedings	Miyamoto K, Huang X, Hashimoto N, Fernandez-Pello C, Fujita O. "Opposed Flame Spread over Polyethylene Insulated Wires Under Varying External Radiation and Oxygen Concentration." 46th International Conference on Environmental Systems, Vienna, Austria, July 10-14, 2016. 46th International Conference on Environmental Systems, Vienna, Austria, July 10-14, 2016. ICES paper ICES-2016-161. , Jul-2016
Papers from Meeting Proceedings	Miyamoto K, Huang X, Hashimoto N, Fernandez-Pello C, Fujita O. "Flammability Limit of Polyethylene Insulated Wires Under Varying Oxygen Concentration and External Radiation." 2016 Spring Technical Meeting of the Western States Section of the Combustion Institute, Seattle, WA, March 21-22, 2016. 2016 Spring Technical Meeting of the Western States Section of the Combustion Institute, Seattle, WA, March 21-22, 2016. , Mar-2016